

Department of Mechanical Engineering

HILSHA BOT

TEAM MEMBERS

AJOY KUMAR SARKER (1810151)

EMON ROY BAPPY (1810158)

MARIFUZZAMAN RIAD (1810166)

SHIHAB KHAN (1810177)

INTRODUCTION

Two-thirds of the earth's surface is covered by oceans and our future can be dramatically changed if we can use the marine resources conveniently. Recently, Our scientific understanding of the ocean is expanding rapidly through the use of a variety of technologies. From this Inspiration, We have manufactured a bio-inspired robotic fish- **Hilsha Bot**. Our main goal is deep-sea scientific exploration & surveillance of the ocean.

CORE SUBSYSTEMS

Buoyancy Control Unit:

- The system consisted of a DC motor, a cylinder, a lead screw, and a spur gear set
- It creates a vacuum inside the cylinder that enables the fish to resurface from deep water.

Tail Actuator:

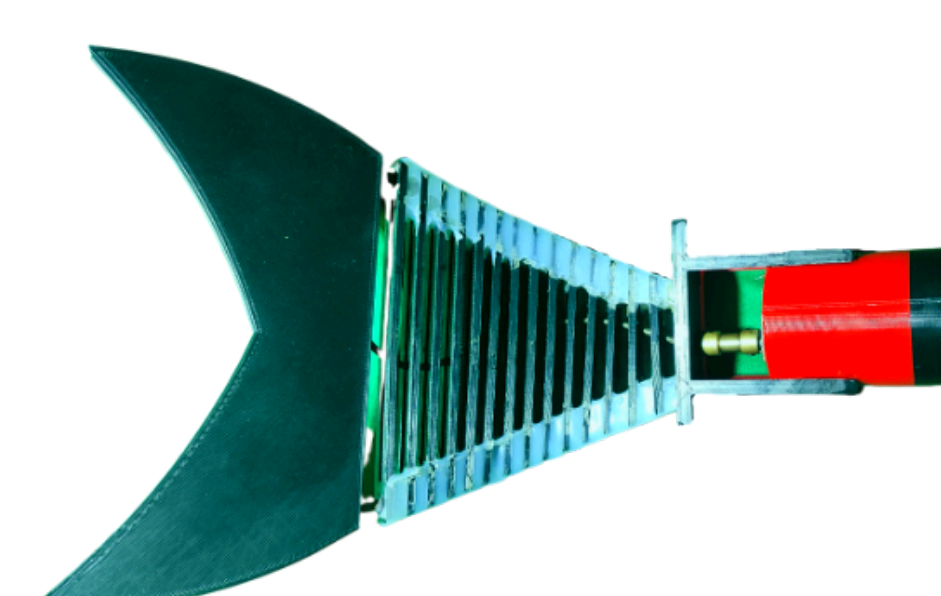
- The tail actuator is made of 3D printed sheets with one slot, a caudal fin, a dc gear motor, a curved-shaped shaft, silicon rubber, and mild steel thin sheet.
- The curve-shaped shaft connected with the motor shaft controls the sheets and creates a sine curve at the tail.

Electronic Systems:

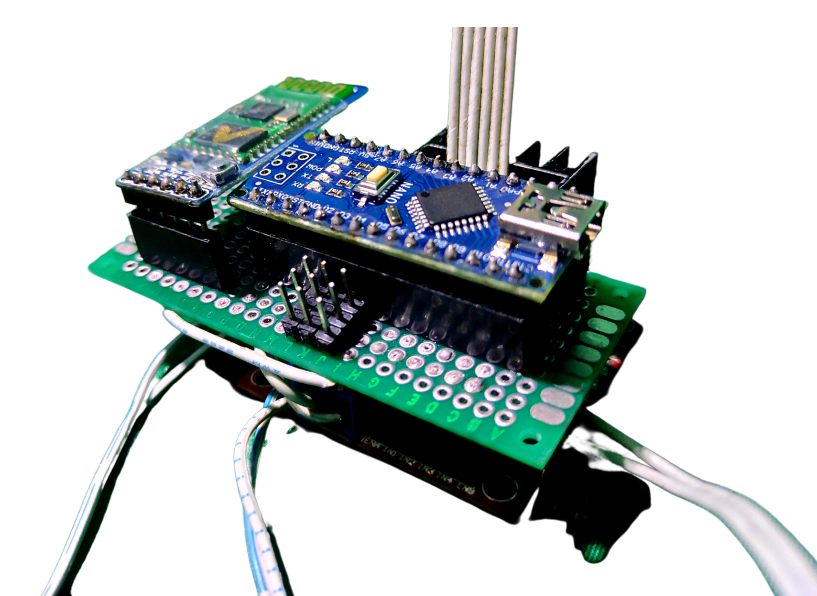
- Arduino Nano Development board is the main processing unit of the system.
- The system controls the actuators through a motor driver and communicates with the operator through Bluetooth communication



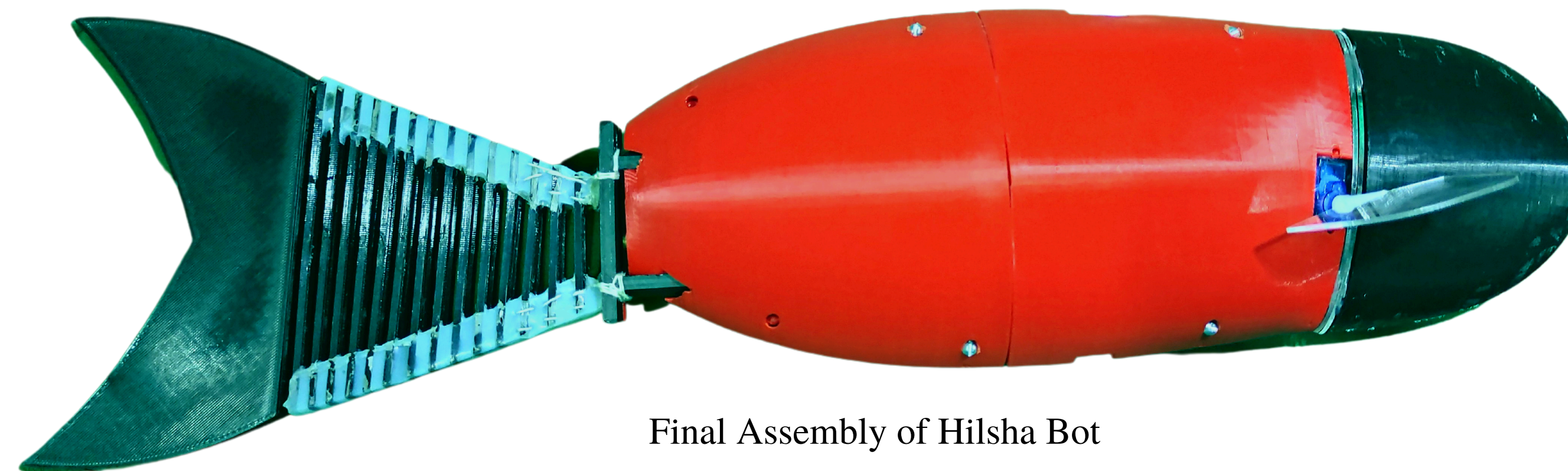
Buoyancy Control Unit



Tail Actuator

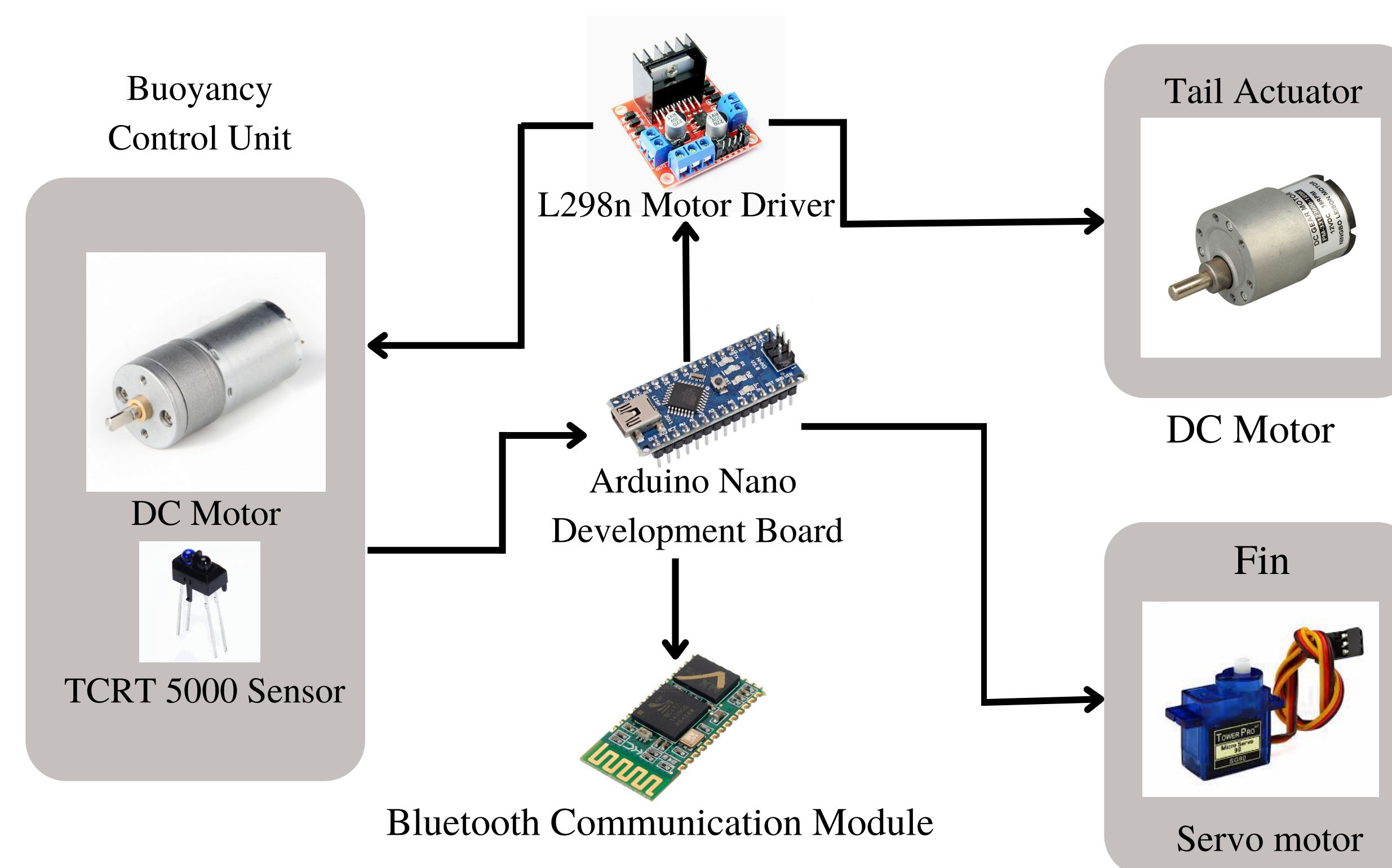


Electronic Subsystems



Final Assembly of Hilsha Bot

SYSTEM ARCHITECTURE



CHALLENGES

- To design buoyancy unit in optimistic length and Not using heavy material.
- Selecting gear design and material.
- Sealing the hole electronic system.
- To build a proper tail actuator system after trial and error method
- Controlling depth using the buoyancy control unit.
- Stability under water
- wireless communication
- unavailability of the motor with proper dimension and torque

APPLICATIONS

- Underwater Investigation and Survey
- Ocean surveillance
- Marin environment explore
- Ocean and marine environment protection
- Military operations especially in the Navy
- Inspection of major infrastructures, such as bridges
- Communication structures

FUTURE PLAN

- Set up a Intel realsense Camera for surveillance
- Modification of our Buyoancy control unit for testing water simultaneously
- Modification of our soft fluidic actuator
- To make a team for its further development & make a another version.

COMPONENTS

- Arduino Nano Development Board
- L298n Motor Driver
- Dc gear motor
- TCRT 5000 Sensor
- Servo
- Bluetooth Communication Module
- Prototype board
- Spur gear set
- Syringe

Supervised By

Dr.Kazi Arafat Rahman, Assistant professor
Shahriar Alam, Lecturer
Priom Das, Lecturer

Project Video

